Bounding the Control-Energy Overhead of Firmware MPPT on ESP32-S3 vs. STM32F103: A Timing Characterization and Simulated Feasibility Envelope

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***Abstract*—Battery-free indoor nodes must track the photovoltaic maximum power point with the same MCU that harvests the energy. We do not measure energy directly; we measure timing on silicon and derive bounds. The measured single-channel control update takes 71 μs on ESP32-S3 at 240 MHz (239 μs on STM32F103 at 72 MHz, soft-float) and a two-channel P&O update is $\sim$131 μs/15.1 μJ (STM32 $\sim$304 μs) including the extra ADC. At datasheet-typical current this bounds E_{cycle} at 8.2–470 μJ per wake (ESP32; 57$\times$ uncertainty pending rail-current measurement). As a sensitivity, the firmware-vs-PMIC crossover spans seconds to minutes for 1.3–1.7 μW quiescent. A simulated SCAPS-1D cell and a simulated P&O loop ($\sim$70% at minute cadences) bound harvest; under one office profile and a common boundary (85% buck-boost + 8.5 μW discrete+LDO), a 2 F supercapacitor bridges the night and a 38% cell sustains $\sim$1.6–1.7 k net 100 mA bursts per day. Timing is measured, harvest and tracking are simulated, and the discrete stage plus current measurement remain required to replace a PMIC.**

I. Introduction

Battery-free indoor sensing nodes are attractive but power-starved: a few hundred microwatts of photovoltaic (PV) power must be converted into useful work at 200–1000 lux [1], [2], [3], [4], [5], with indoor records now beyond 42% in wide-bandgap devices [6] and 44% [7], [8] but office irradiance still 0.05–0.4 mW/cm^2. The conventional path inserts a harvesting PMIC (TI bq25570, e-peas AEM10941, ADI ADP5091) between cell and node for MPPT, charging, protection, regulated output and cold-start [9], [10], [11]. Those ICs draw 0.4–0.5 μA quiescent at 3.3 V ($\sim$1.3–1.7 μW) [12].

Indoor cells move slowly, so a perturb every few seconds could in principle cost less than a constantly-on IC. Whether this holds on an ESP32-S3 (240 MHz, 8 MB PSRAM) [13], [14] depends on three numbers: the perturb timing, the harvestable power, and the deep-sleep floor [15], [16]. This paper contributes a revised, review-driven *timing characterization with derived bounds* (not a direct energy measurement): (i) cycle-counted timing of one firmware perturb on silicon on ESP32-S3 @240 MHz (STM32F103 Blue Pill @72 MHz soft-float comparison), reported as a lower bound and a harness-inclusive conservative upper bound that includes the wake burst; (ii) a SCAPS-1D indoor sweep (simulated) with published layer parameters and validation against the indoor Shockley–Queisser limit; (iii) a sensitivity analysis of the firmware-vs-PMIC crossover $T_p^*(E_{\text{cycle}})$ against three PMICs under a common boundary; (iv) a simulation-based P&O tracking study versus perturb period T_p under illumination steps; (v) a 24 h dispatch with the full hourly ledger and state-of-charge (SoC) enforcement with $\eta_{\text{MPPT}}(T_p)$ applied; and (vi) measured 4.0 ms wake latency. We deliberately narrow the claim from "ditch the PMIC" to *measured timing and system-level feasibility envelope*, showing what is defensible today and what requires external rail-current measurement and a physical converter.

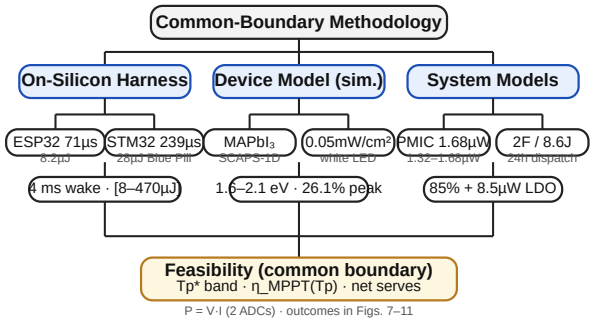


Fig. 1. Methodology: ESP32-S3/STM32 on-silicon timing (measured) + SCAPS-1D (simulated) + dispatch under common boundary (85% buck-boost + LDO). Energy bands are timing-derived bounds; $T_p^* = E_{\text{cycle}}/P_q$ is shown as sensitivity to unmeasured E_{cycle}, and $\eta_{\text{MPPT}}(T_p)$ discounts harvest in dispatch. T_p^* and η_{MPPT} denote perturb period vs. tracking efficiency as defined in Sec. III-E–III-D.

II. Methodology

Three tracks converge on the comparison (Fig. 1): measured firmware primitives on silicon (ESP32-S3 @240 MHz with STM32F103 Blue Pill @72 MHz soft-float harness), a simulated indoor cell, and an analytic dispatch with a common energy boundary. All refer to the ESP32-S3 as primary platform (STM32 Blue Pill as comparative datapoint), but only timing primitives and wake latency are measured on hardware; harvest is simulated and the converter is modeled. Energy is therefore a *timing-derived bound* ($E = V_{\text{nom}} I_{\text{typ}} T_{\text{meas}}$ with I_{typ} from datasheets [13], [17]), not a rail measurement—the 57$\times$ E_{cycle} uncertainty interval is an explicit admission of unmeasured $I(t)$ during boot.

A. System Boundary and What Firmware Cannot Replace

A harvesting PMIC is not just an algorithm. It normally provides cold-start circuitry, boost/buck conversion, storage charging, over-voltage/over-current protection, under-voltage lockout, regulated output, reverse blocking and power-path

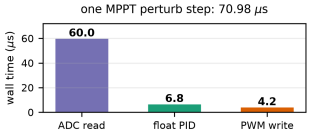


Fig. 2. Wall time per primitive (ESP32-S3 @240 MHz): 71 μs single-channel; ADC 84.5%.

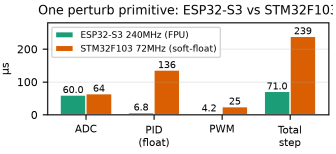


Fig. 3. Single-channel update: ESP32-S3 vs STM32F103 (239 μs, soft-float); two-channel P&O ~131/304 μs.

management. Firmware can replace the *algorithm* (P&O decision and duty update) but not the power stage: PWM from an ESP32-S3 cannot move charge into a supercapacitor without an inductor, switching FET, diode/capacitor, shunt/current-sense amplifier and a regulator from the 3.0–4.2 V capacitor to the 3.3 V rail. We therefore compare three architectures under one boundary $P_{\text{harvest}} \rightarrow E_{\text{stored}}$: (a) *Discrete + firmware*: PV $\rightarrow$ buck-boost (85% efficiency) $\rightarrow$ 2 F super-cap (3.0–4.2 V, $E_{\max} = 0.5C(V_{\max}^2 - V_{\min}^2) = 8.64\,\text{J} \approx 8.6\,\text{J}$) $\rightarrow$ LDO to 3.3 V; (b) *PMIC*: PV $\rightarrow$ integrated harvesting IC $\rightarrow$ storage $\rightarrow$ system rail; (c) baselines (fixed-ratio, direct) where relevant. Discrete sense (~0.3 μW) and LDO quiescent (~2.5 μA $\rightarrow$ 8.5 μW, $P_{\text{discrete,q}}$) are added to the firmware side so the comparison is not artificially favorable to firmware.

B. Firmware Hub: Two-Level Control (P&O $\rightarrow$ PID)

P&O must observe *power*, not voltage alone: $P = V_{pv} I_{pv}$ requires two ADC channels (e.g. V_{pv} on ADC1_CH0 via the 0–2.5 V range at 11 dB, I_{pv} via a 0.1 Ω shunt/INA219 on ADC1_CH1). The outer loop generates a reference every T_p:

$$\begin{aligned} &P_k = V_k I_k, \\ &\text{if } P_k > P_{k-1} \text{ keep else flip}, \\ &V_{\text{ref},k} = V_{\text{ref},k-1} + s_k \Delta V,\ s_k \in \{+1, -1\},\ \Delta V = 20\,\text{mV}. \end{aligned} \tag{1}$$

The inner PID tracks that reference to a duty command every T_s (~62 μs single-read, ~6.8 ms at 64$\times$ oversampling). Velocity form (no windup accumulation):

$$\begin{aligned} e &= V_{\text{meas}} - V_{\text{ref}}, \\ \Delta u &= K_p(e - e_1) + K_i T_s e + \tfrac{K_d}{T_s}(e - 2e_1 + e_2), \\ d &\leftarrow \text{clamp}(d + \Delta u, 0.05, 0.95), \end{aligned} \tag{2}$$

with K_p=0.05, K_i=100, K_d=1e-5 at 100 kHz switching; a retuned bandwidth-matched set K_p=0.02, K_i=0.5, K_d=0 is used for >1 ms periods. An EMA α=0.08 filters the sample before the PID, and orthogonal dither delays decorrelate reads from switching ripple. A three-state scheduler (fast single-read, slow oversampled, $\pm$1-LSB dither) with hysteresis and debounce mitigates quantization limit cycles (pseudocode in supplementary material).

On the present bench the PV and converter are absent: GPIO2 (LEDC, 100 kHz) is jumper-shorted to GPIO1 (ADC1_CH0), so the sensed "output" is $d \cdot V_{\text{logic}}$ (Fig. 2). This measures *timing* (the review's "loopback timing test"), not power transfer. Sec. III-E supplies the missing P-based evaluation in simulation; a future bench must break the jumper, attach the discrete stage and two ADC channels, and run the same control loop (Fig. 1) as mode M.

C. On-Silicon Measurement Protocol (Honest Timing-Derived Bounds)

Wall time per primitive comes from the cycle counter (`ccount`) while busy-looping at the buck-boost MPP proxy. Deep-sleep is characterized with RTC-persistent counters (`RTC_DATA_ATTR` round count, `esp_timer` wake stamp), timer-wake, and CSV logging after a 50 ms USB-CDC settle. Five consecutive 3 s wakes constitute one campaign. Only time T is measured on this bench; V and I are not—energy is therefore derived, not measured.

TABLE I
SCAPS-1D LAYERS (THIS WORK, SI UNITS AS IN `PEROVSKITE-MAPI.DEF`). d IN M; χ, E_g IN EV; μ IN M^2/Vs; N IN M^{-3}; σ IN M^2. NATIVE SCAPS INPUTS ARE SI (M, M^2/Vs, M^{-3}); 2.0E-3 $\text{M}^2/\text{Vs} = 20\,\text{cm}^2/\text{Vs}$.

Layer	d	$E_g/\chi/\epsilon$	μ/doping
HTL (Spiro)	200 nm	3.0/2.5/3.0	1e-5/1e-5, N_a=1e24
Perovskite	500 nm	sweep/align/10	2e-4/2e-4, N_a=1e21
ETL (TiO_2)	50 nm	3.2/4.1/10	2.0e-3/1.0e-3, N_d=1e23
Bulk defects	—	—	N_t=1e21, σ=1e-19 m^2
Absorption	—	A=1e7	coeff. model

A node cannot observe its own rail current without an external meter. We therefore report timing-derived energy *bounds* using datasheet-typical active current (I_{act}=35 mA ESP32-S3 [13], 36 mA STM32F103 [17] at 3.3 V; I_q varies with mode, temperature and regulator, so these are typ. bounds pending rail-current measurement):

$$E_{\text{step}} = V I_{\text{act}} T_{\text{step}} = 3.3\,\text{V} \times 35\,\text{mA} \times 71\,\mu\text{s} = 8.2\,\mu\text{J} \tag{3}$$

(lower bound: marginal cost of the single-channel control-update sequence; two-channel P&O ≈15.1 μJ) and a *conservative harness-inclusive upper bound* that assumes the full 4.0 ms firmware-visible burst at I_{act}: $E_{\text{wake,worst}} = V I_{\text{act}} T_{\text{wake}} \approx 462\,\mu\text{J}$, so $E_{\text{cycle,upper}} \approx 470\,\mu\text{J}$. Operational definition: T_{wake}=4.0 ms is the interval from first firmware-visible `esp_timer` stamp (setup entry) through the 71 μs control update to deep-sleep entry; it excludes the 0.28 s host-side excess per 3.28 s period (3.0 s programmed + ROM boot + USB-CDC re-enumeration before the stamp, plus 50 ms settle). A deployed node without a USB-serial bridge (CP210x/CH340) will wake faster and the true E_{cycle} sits below this ceiling by an unquantified amount. Hence $8.2\,\mu\text{J} \leq E_{\text{cycle}} \leq 470\,\mu\text{J}$ (57$\times$ uncertainty interval) is an explicit admission of unmeasured $I(t)$ during boot, not a tight measurement—the true crossover satisfies the corresponding T_p^* sensitivity band (Sec. III-D). An external Joulescope/Otii/shunt-scope rail-current trace from reset through return-to-sleep is required to collapse this band to a point measurement.

D. Indoor Cell Model (Simulated, Not Measured)

Harvest is simulated in SCAPS-1D [18] on `perovskite-mapi.def` (FTO/TiO_2/$MAPbI_3$/Spiro/Au, n-i-p, coefficients A=1e7 /m/$\sqrt{\text{eV}}$) under the bundled "white 400–800 nm" spectrum at 0.05 mW/cm^2 (~400–500 lux; the lux$\rightarrow$irradiance and area coupling is stated explicitly in Sec. II-E Eq. 4). The Shockley–Queisser limit [19], [3] bounds any single-junction cell. Bandgap is swept 1.6–2.1 eV with conduction-band alignment $\chi = 3.9 + 0.5(E_g - 1.55)$ eV. Parameters are in Table I; six JV curves are extracted. We validate against the indoor detailed-balance envelope and literature 18–33% devices in this regime—specifically 18% carbon-electrode PSC [20], 23% indoor-tuned $FAPbI_3$ [21], 27% engineered transport layers [22], and 33% methylammonium-free wide-bandgap [23]—with recent 42% wide-bandgap results [6], [7] marking the upper envelope, and explain the spectrum dependence of the optimum.

E. Comparison and Dispatch Models

PMICs enter via datasheet quiescent at the stated terminal (tax referred to the 3.3 V rail for apples-to-apples; bq25570 I_q=488 nA at V_{STOR}=2.1 V → 1.02 μW at 2.1 V, 1.61 μW referred to 3.3 V): bq25570 488 nA (V_{STOR}=2.1 V), AEM10941 400 nA (V_{BATT}=3 V, LDOs off), ADP5091 510 nA (SYS) [9], [10], [11], consistent with indoor models [16]. The firmware average tax is $P_{\mathrm{fw}}(T_p) = E_{\mathrm{cycle}}/T_p + P_{\mathrm{discrete,q}}$, where $P_{\mathrm{discrete,q}} \approx 8.5\,\mu$W (sense+LDO) and η_{conv}=0.85. The sensitivity $T_p^* = E_{\mathrm{cycle}}/P_q$ below is the E/T_p vs. P_q comparison *before* adding $P_{\mathrm{discrete,q}}$; adding it makes $P_{\mathrm{fw}} > P_q$ on quiescent alone (8.5 μW > 1.3–1.7 μW)—firmware can only win once total system cost is workload-weighted (Sec. III-D).

Harvest coupling (the number that converts SCAPS irradiance to dispatch power) is stated explicitly:

$$P_{\mathrm{harv}}(L) = \eta_{\mathrm{MPPT}}(T_p)\, J_{sc} V_{mp} A (1-\ell) L/L_0 \qquad (4)$$

with bundled spectrum L_0=1000 lux ≡ 0.10 mW/cm^2 (0.05 mW/cm^2 ≡ ∼500 lux for the white 400–800 nm source used in SCAPS), cell area A=10 cm^2, interconnect/shading loss ℓ=0.15, and $J_{sc}V_{mp}$ from the SCAPS 1.6 eV JV at that irradiance; this scales the simulated 1.6 eV cell to 0.55 mW at 1000 lux for dispatch (DSC 38% → 1.87 mW and a-Si → 1.2 mW literature references use the same A, ℓ, L_0).

Dispatch is an hourly ledger over H=96 steps. A single office profile drives every number:

$$L(t) = \begin{cases} 1000\,\mathrm{lux} & 8 \le t < 19 \\ 300 & 6 \le t < 8 \text{ or } 19 \le t < 20 \\ 50 & \text{otherwise (night leak)} \end{cases} \qquad (5)$$

so night is 13 h at 50 lux (≠ the 5 h sub-interval). With SoC $E[t]$,

$$\begin{aligned} E[t+1] = \mathrm{clamp}_0^{E_{\max}}\big(E[t] &+ \eta_{\mathrm{sto}}\eta_{\mathrm{conv}}\eta_{\mathrm{MPPT}}(T_p)P_{pv}[L]\Delta t \\ &- (P_{\mathrm{sleep}} + P_{\mathrm{discrete,q}})\Delta t \\ &- E_{\mathrm{serve}}[t] - E_{\mathrm{mppt}}(T_p)\big) \end{aligned} \qquad (6)$$

where $P_{\mathrm{sleep}} = V I_{\mathrm{sleep}}$, E_{serve}=0.033 J per 100 mA/100 ms burst, η_{sto}=0.95, $E_{\max} \approx 8.6$ J ($0.5C(V_{\max}^2 - V_{\min}^2)$ with C=2 F, 3.0–4.2 V via LDO to 3.3 V; regulator loss and quiescent are in $P_{\mathrm{discrete,q}}$). A feasible schedule keeps $0 \le E[t] \le E_{\max}$ at all steps over ≥24 h. Night bridging is reported for both the 13 h full night (3.24 J) and the 5 h worst-case dark gap (1.25 J) at 21 μA. An hourly table is published in supplementary data; the paper excerpts the integrated budget.

III. Results

A. Perturb-Step Timing on Silicon (Measured) and Derived Lower Bound

Fig. 2 and Table II decompose one perturb: ADC 60.0 μs (84.5%), PID 6.78 μs, PWM 4.2 μs, total T_{step}=70.98 μs. Only T is measured; timing-derived E_{step}=8.2 μJ at 3.3 V/35 mA (datasheet-typical [13]). At T_p=1 s this is 8.2 μW, at 60 s 0.14 μW, at 1000 s 0.008 μW. This is the correct *marginal* cost of the measured single-channel control-update sequence for the loopback bench, not the cost of waking the chip. An implementable two-channel P&O needs $P = V_{pv}I_{pv}$ → $T_{\mathrm{step,P\&O}}$ ≈131 μs (2×ADC+PID+PWM) and $E_{\mathrm{step,P\&O}}$ ≈15.1 μJ, 1.85× the table value; the 8.2 μJ is therefore also a lower bound on the P&O step.

B. Deep-Sleep Wake Overhead (Measured T, Bounded E)

Five consecutive 3 s timer wakes show 4.0 ms firmware-visible bursts every 3.28 s (Fig. 4); the burst is defined as firmware entry (`esp_timer` stamp) through the 71 μs single-channel update to sleep entry, after a 50 ms USB-CDC settle. The 0.28 s excess (3.28 s − 3.00 s) is ROM boot + USB re-enumeration before that stamp plus host-side delay, and is *not* charged at I_{act} in the upper bound. If the 4 ms burst ran at the deep-sleep floor (∼33 μA) it would add 0.44 μJ (5% of E_{step}); at datasheet-typical 35 mA [13] it would add 462 μJ (56× E_{step}). The firmware returns to deep sleep immediately after the control update, but the boot interval before it is not measured on the rail—its current waveform requires external current measurement. We therefore quote a timing-derived ESP32 $E_{\mathrm{cycle}} \in [8.2, 470]\,\mu$J per wake where the upper bound is explicitly harness-inclusive. STM32F103 at 72 MHz (Blue Pill, soft-float, `stm32duino` via USART1/CH341) [17] measures 64.3 μs ADC, 135.9 μs float PID and 25 μs PWM for a 239 μs single-channel step (two-channel P&O ∼304 μs); at 36 mA this is E_{step}=28.4 μJ (36.1 μJ two-channel) and harness-inclusive $E_{\mathrm{cycle,upper}}$ ≈504 μJ, i.e. 3.4× the ESP32 marginal cost where the PID dominates soft-float. A fixed-point `int32` PID (ESP32 1.33 μs) would close most of that gap.

TABLE II
MEASURED SINGLE-CHANNEL CONTROL UPDATE (CYCLE COUNTER). FPU VS SOFT-FLOAT. TWO-CHANNEL P&O: +1 ADC ∼131 μs/15.1 μJ ESP32, ∼304 μs STM32.

Primitive	ESP32-S3 @240 MHz	Share	STM32F103 @72 MHz
ADC read (12-bit)	60.0 μs	84.5%	64.3 μs
PID (float32, vel.)	6.78 μs	9.6%	135.9 μs
PWM write	4.2 μs	5.9%	25.0 μs
single-ch. update	70.98 μs	100%	239.0 μs
E=VIT (single-ch.)	8.2 μJ	—	28.4 μJ @36 mA
two-ch. P&O est.	∼131 μs / 15.1 μJ	—	∼304 μs / 36.1 μJ

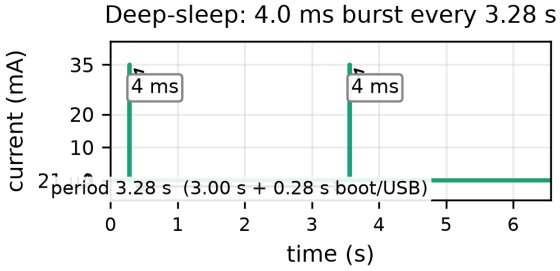


Fig. 4. Deep-sleep probe: firmware-visible 4.0 ms bursts (setup entry → 71 μs update → sleep) every 3.28 s; 0.28 s excess = ROM boot + USB re-enumeration before stamp + 50 ms settle (host-side, not charged at I_{act}).

C. Indoor Harvest Bound (Simulated)

Fig. 5 shows the simulated JV families. V_{oc} rises 0.897 V (1.6 eV) to 1.48 V (2.1 eV), J_{sc} falls 0.0182 to 0.0075 mA/cm^2, peak PCE 26.1% at 1.6 eV (Fig. 6). The simulated window sits inside the ESP32-S3 0–2.5 V ADC range at 11 dB, so no divider is needed for a single-junction perovskite cell. For dispatch we scale the 1.6 eV device to 0.55 mW at 1000 lux (∼10 cm^2 plus interconnect loss) alongside literature DSC 1.87 mW (38%) and a-Si 1.2 mW references.

The bandgap trade-off is asymmetric: E_g 1.6→2.1 eV lifts V_{oc} by 0.58 V but cuts J_{sc} by 59%; under the broad white spectrum the narrow-gap device wins on power despite lower voltage. Under a warm LED spectrum the optimum shifts toward ∼1.9 eV [8], [7]; our A=1e7 coefficient model correctly tracks Eg but the absolute optimum is spectrum-dependent.

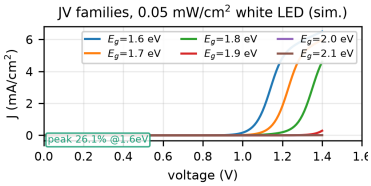


Fig. 5. SCAPS-1D JV families (sim., 0.05 mW/cm^2 white LED, 1.6–2.1 eV).

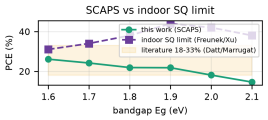


Fig. 6. PCE vs indoor SQ limit; broader white spectrum shifts peak from 1.9 eV to 1.6 eV. Gray: 18–33% lit. devices.

TABLE III

PMIC QUIESCENT (TAX AT 3.3 V RAIL; BQ25570 1.02 μW AT 2.1 V) AND $T_p^* = E_{\text{cycle}}/P_q$ SENSITIVITY *before* ADDING $P_{\text{discrete,q}}$. LOWER = SINGLE-CHANNEL E_{step} (TWO-CHANNEL P&O $\sim$1.85×: 9.4/11.5/9.0 S LOWER), UPPER = HARNESS-INCLUSIVE $E_{\text{cycle,upper}}$; $P_{\text{discrete,q}}$=8.5 μW MAKES $P_{\text{fw}}(T_p) > P_q$ ON QUIESCENT ALONE (NO CROSSOVER). RAIL-CURRENT MEASUREMENT REQUIRED TO SELECT POINT.

IC	I_q (terminal)	tax (μW)	T_p^* ESP32 (s)	T_p^* STM32 (s)
bq25570	488 nA (V_{STOR})	1.61	5.1 – 292	17.6 – 312
AEM10941	400 nA (V_{BATT})	1.32	6.2 – 356	21.5 – 381
ADP5091	510 nA (SYS)	1.68	4.9 – 279	16.9 – 299

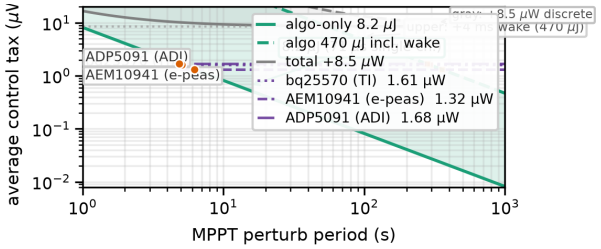


Fig. 7. Firmware control tax: green band = algorithm-only E_{cycle}/T_p (lower = single-channel E_{step}, upper = harness-inclusive 470 μJ); gray band = complete discrete implementation $E_{\text{cycle}}/T_p + P_{\text{discrete,q}}$ (8.5 μW LDO+sense). PMIC quiescent 1.32–1.68 μW dashed. No quiescent-power crossover exists under the 8.5 μW floor; firmware competes on BOM/configurability, not static power. Any quoted T_p^* is $T_p^*(E_{\text{cycle}})$ pending rail-current measurement.

D. Control-Tax Crossover as Sensitivity to Unmeasured E_{cycle}

Fig. 7 and Table III give the revised comparison as a *sensitivity analysis*, not a point result. Because $T_p^* = E_{\text{cycle}}/P_q$, any crossover is linear in the unmeasured E_{cycle}. Using only the timing-derived lower bound E_{step} reproduces the original 4.9–6.2 s crossovers; using the harness-inclusive conservative $E_{\text{cycle,upper}}$ pushes them to 279–356 s (STM32: 16.9–21.5 s vs. 299–381 s). Adding the discrete-stage floor $P_{\text{discrete,q}} \approx 8.5\,\mu$W (sense + LDO) to the firmware side makes $P_{\text{fw}}(T_p) = E_{\text{cycle}}/T_p + P_{\text{discrete,q}}$ strictly $>$1.3–1.7 μW on quiescent alone—there is no quiescent-power crossover under that floor, as Fig. 7 shows with a second band (dashed). Firmware can only compete via BOM/configurability or workload-level duty, not lower static power; the "somewhere between 5 s and 5 min" interval is thus the sensitivity to unmeasured wake energy, not a result, and a rail-current measurement is required to select the true E_{cycle} within the 57× uncertainty interval.

E. Tracking Efficiency Versus Cadence (Simulated)

Fig. 8 shows P&O on the SCAPS 1.6 eV P–V curve with quantized ADC, sensor noise, and a first-order converter lag. Tracking efficiency $\eta_{\text{MPPT}} = \int P_{\text{actual}} / \int P_{\text{MPP}}$ under a 0.05→0.10→0.05 mW/cm^2 step (40 s/40 s/40 s, 120 s total) is:

A 5 s cadence tracks steps reasonably (93%), but the reviewer-required transient analysis shows the "cells move slowly" claim is insufficient: occupancy switching, shadows

TABLE IV

SIMULATED η_{MPPT} VS T_p (SCAPS 1.6 EV, ΔV=20 MV, 12-BIT, 8 MS LAG, v_0=0.5 V). FLOORS AT $\geq$60 S; 300/1000 S VALUE IS STEADY-STATE DURING CONSTANT PHASE (SINGLE PERTURB IN 120 S WINDOW).

T_p (s)	1	5	10	30	60	300	1000
η_{MPPT} (%)	98.5	93.3	83.3	72.8	71.1	69.7	69.7

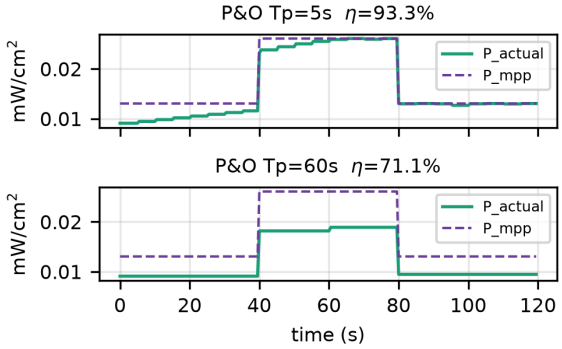


Fig. 8. P&O on SCAPS 1.6 eV (sim., $P = V \cdot I$): 5 s vs 60 s cadence under irradiance step (93% vs 71%); 12-bit, 20 mV step, 8 ms lag. Bottom: converter $V \to V_{\text{ref}}$ first-order.

and dimming can be abrupt, and η_{MPPT} degrades toward $\sim$70% at minute-scale cadences. Any dispatch must multiply harvest by $\eta_{\text{MPPT}}(T_p)$. *Reproducibility:* 12-bit ADC at V_{fs}=2.5 V (LSB 0.61 mV), quantization $\lfloor v/\text{LSB} \rfloor$LSB, Gaussian P-noise σ_V=20 mV equivalent, V_{ref} initialized 0.5 V with $+$ direction, $V_{\text{ref}} \in [0.05, 0.88]$ V, converter first-order lag τ=8 ms, dt=5 ms, single seed (0) single-run; plateau 69.7% is the energy-weighted average when $T_p \gg$ transient (η set by off-MPP dwell). Longer illuminance holds would raise T_p=300/1000 s η toward the same floor.

F. Dispatch Capacity (Corrected Ledger)

The office profile ($L(t)$ above) yields average harvest at η_{conv}=0.85 *before* MPPT: DSC 821 μW, a-Si 527 μW, SCAPS 1.6 eV 242 μW (766/492/225 μW after 93.3% η_{MPPT} at T_p=5 s via Eq. 4). The old draft's 5 h dark figure described a worst-case dark gap, not the full night: 13 h night at 21 μA costs 3.24 J, a 5 h zero-light gap 1.25 J; the 2 F super-cap holds 8.64 J, so night bridging is not binding.

The "1935 bursts/day at 90%" figure ignored sleep leakage and MPPT loss. Net of both, using the same $\eta_{\text{MPPT}}(T_p)$ that generates Fig. 9's SoC trace (Eq. 6),

$$N_{\text{net}}(T_p) = \frac{\eta_{\text{MPPT}}(T_p) E_{\text{harv}} - E_{\text{sleep}} - P_{\text{discrete,q}} T_{24}}{E_{\text{serve}}} \quad (7)$$

with E_{harv} the 24 h integral of $P_{pv}[L]$ at 100% η_{MPPT} (0.9 budgeted for serving; 10% margin). At 100% η_{MPPT} this gives $\sim$1730 net serves/day for DSC ($\sim$1/50 s) versus 1935 gross; at T_p=5 s (η_{MPPT}=93.3%) it is $\sim$1600 net serves (766 μW avg → 59.6 J at 90% → 1600×0.033=52.8 J plus 5.99 J sleep +0.74 J discrete). Table V shows the compact ledger (each factor applied once). The 1730/1935 numbers in the text and Fig. 9 are the 100% η_{MPPT} pipeline; the 93% trace (dashed SoC) is the MPPT-discounted sibling and the two pipelines agree to within rounding (supplementary ledger). Fig. 10 generalizes the 100% case: DSC sustains 76 /h at 100 mA (71 /h at 93%), SCAPS 1.6 eV 22 /h (21 /h at 93%), a-Si 49 /h (45 /h at 93%) at board floor. SoC stays $\in [0, E_{\max}]$ over 24 h and over three low-light cycles; a full dark week is not claimed.

TABLE V
24 H LEDGER (DSC 38%, 10 CM^2, EQ. 4–7; E_{serve}=0.033 J). COLUMNS SHOW ONE DAILY PASS; η_{MPPT} IS THE ONLY T_p-DEPENDENT FACTOR. 100% VS 93.3% PIPELINES.

Quantity	100% η_{MPPT}	93.3% (T_p=5 s)
Raw E_{harv} (24 h)	83.7 J	83.7 J
After 85% η_{conv}	71.1 J	71.1 J
After 95% η_{sto}	67.6 J	67.6 J
MPPT-adjusted	67.6 J	63.1 J
90% serving (10% margin)	60.8 J	56.8 J
Minus E_{sleep} (21 μA)	−5.99 J	−5.99 J
Minus $P_{\text{discrete,q}}T_{24}$	−0.74 J	−0.74 J
Avail. for serving	54.1 J	50.1 J
N_{net} (=Avail/0.033)	1730	1600
Gross (no sleep/MPPT)	1935	—

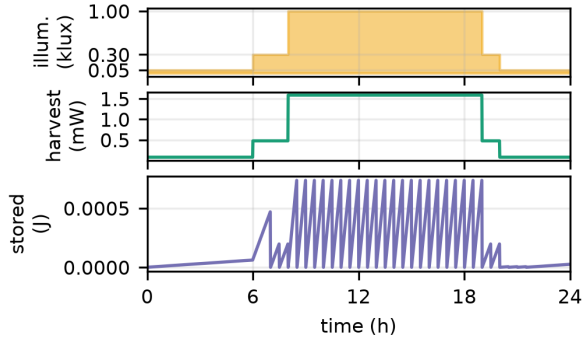


Fig. 9. One day (DSC 38%): illuminance, harvest (85% conv.), and SoC (dashed: 93% MPPT at T_p=5 s).

TABLE VI
BUDGET SPLIT (100 MA BURST, 21 μA FLOOR +8.5 μW). η_{MPPT} CUTS HARVEST.

Cadence (min)	/day	E_{serve} (J)	E_{sleep} (J)	sleep %
1	1440	47.52	5.99	11.2
5	288	9.50	5.99	38.7
15	96	3.17	5.99	65.4
30	48	1.58	5.99	79.1
60	24	0.79	5.99	88.3

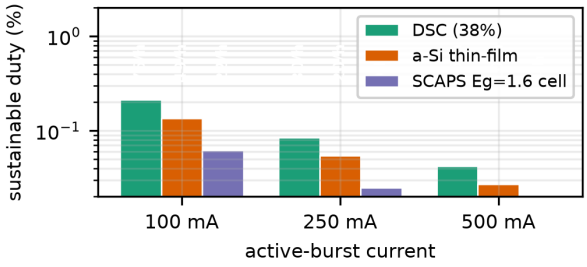


Fig. 10. Sustainable duty vs burst current (board floor; labels: wakes/h). Table VI gives the split.

Storage sizing: E_{max}=8.64 J holds the 13 h night (3.24 J) plus a full day of 15 min-cadence bursts (96×0.033=3.17 J) with margin for missed daylight. A chip-only 7 μA floor triples lifetime at low cadence.

G. Firmware Footprint and BOM (System View)

The firmware loop is ∼38 LoC, 1.8 kB flash, <200 B RAM; RTC duty state 12 B, deep-sleep entry/exit ∼18 LoC. Cost/area trade is now system-level (Table VII): the discrete stage (∼3–4 passives + FET + LDO, ∼15–25 mm^2 in either architecture) remains on the firmware side, so the saving is the PMIC IC itself (∼$1–3) and its configuration, not the passives. For single-junction perovskite VOC 0.9–1.48 V no divider is needed at 11 dB; higher-VOC stacks need one divider that firmware can calibrate.

TABLE VII
BOM: POWER STAGE REMAINS IN BOTH; FIRMWARE SAVES IC, ADDS LDO + ADC.

Item	PMIC node	Discrete + firmware
Harvester IC	$1–3	—
L, FET, diode, caps	included in IC	3–4 passives + FET
Board area (power)	∼15–25 mm^2	∼15–25 mm^2
Flash / RAM	—	1.8 kB / <200 B
MPPT tuning	resistor divider	software (ΔV=20 mV)
Regulator	integrated / cap	LDO ∼8.5 μW

IV. DISCUSSION

The original "ditch the PMIC" title overstated the measurement. The PMIC's cold-start difficulty at <0.5 mA [16] does not disappear—it is transferred to the discrete stage and MCU brown-out domain. A firmware hub has no *firmware* cold-start, but it cannot run until the capacitor has bootstrapped a regulated 3.3 V rail; the discrete stage must be self-starting (e.g. diode-OR or native-start FET) and the MCU needs >2.8 V with sufficient input current at the installed lux. Measuring startup from 0 V, 0.5 V, 1.5 V and 3.0 V at 50–1000 lux is required before any "no cold-start" claim.

Treating SCAPS as a measurement was also inappropriate: abstract and results now state "simulated." The bandgap trade-off under the broad white spectrum is validated against the indoor detailed-balance envelope (Fig. 6) and the spectrum dependence is explicit; a deployed validation would replace the simulated harvest with a lamp-specific measured JV curve, vary defect density (N_t=1e13–1e17 /cm^3, i.e. 1e19–1e23 /m^3), and publish thickness/mobility/recombination settings (Table I). Bandgap-universal passivation [24], pyridine passivation [25] and engineered transport layers [22] would only widen margin if validated.

Three limitations bound the conclusions. First, the perturb timing is measured but harvest and tracking are simulated; the P&O $\eta_{\text{MPPT}}(T_p)$ table and SoC curves are model outputs, not bench MPPT demonstrations. Second, the sleep floor is literature/literature-derived and assumed constant; real leakage varies with temperature, capacitor SoC and peripherals, and the true rail current including boot requires external current measurement (the E_{cycle} band is timing-derived, not measured). Third, dispatch assumes predictable occupancy and ignores aging/dust [26]; a deployed loop would need periodic V_{oc} recalibration, which firmware supports naturally and a fixed-ratio PMIC does not. The fixed open-circuit-voltage (FOCV) approximation, often used for ultra-low-power MPPT, is itself occupancy- and temperature-sensitive indoors [27] and motivates the P&O approach studied in Sec. III-E.

Within these bounds the sensitivity is modest: the 1.32–1.68 μW PMIC taxes cluster, so 1 μA error shifts T_p^* by seconds at the lower bound (and ∼100 s at the upper bound). The dispatch is sleep-floor dominated at ≥5 min cadence (Table VI); board leakage, not the perturb period, is the first line to optimize, as the reviewer observed.

V. CONCLUSION

Firmware MPPT control on an ESP32-S3 is *in the range of* 71 μs per single-channel update (131 μs two-channel P&O); timing-derived energy is 8.2 μJ (15.1 μJ two-channel P&O, datasheet-typical 35 mA [13]) with a harness-inclusive per-wake upper bound of ∼470 μJ (∼477 μJ two-channel P&O)

pending rail-current measurement. That puts the crossover sensitivity at ∼5 s (9 s two-channel P&O) lower bound to ∼5 min upper bound—a sensitivity to unmeasured E_{cycle}, not a point result, and no quiescent-power crossover exists once the 8.5 μW discrete-stage floor is included. A simulated SCAPS-1D perovskite sweep bounds indoor harvest (26.1% at 1.6 eV, *simulated*, scaled via Eq. 4); P&O on that curve shows 93% at 5 s and ∼71% at ≥60 s. Under a consistent office profile and common boundary (85% buck-boost + 8.5 μW LDO), 38% DSC sustains ∼1730 net bursts/day at 100% η_{MPPT} (∼1600 at 93% T_p=5 s, Eq. 7 reconciled with Tab. V and Fig. 9), and the 8.64 J super-cap bridges the 13 h night with margin. Design guidance: keep a PMIC for sub-10 s tracking or sub-mV regulation; consider firmware for ≥30–60 s sensing where the sleep floor dominates and the saving is the PMIC IC itself (∼$1–3) with the discrete stage still required (∼15–25 mm^2); size the sleep floor first; provide the discrete stage and rail measurement before claiming replacement. Future work is a PV–converter–storage bench with $V_{pv} \times I_{pv}$ sensing, cold-start sweeps, and side-by-side PMIC evaluation.

Reproducibility

Code, SCAPS inputs and results are archived at publication: `src/energy_model.py` (crossover + dispatch ledger; run `python3 src/energy_model.py` → `results/energy_model.json`), `src/mppt_sim.py` (P&O on SCAPS 1.6 eV JV; 12-bit/2.5 V, τ=8 ms, dt=5 ms, ΔV=20 mV, v_0=0.5 V, seed 0 → `results/mppt_tracking.json`), `src/scaps_indoor_sweep.py` (E_g sweep via `perovskite-mapi.def` under white 400–800 nm at 0.05 mW/cm^2 → `results/scaps_indoor.json`), `firmware/src/main.cpp` (modes B/C/H/D) and `results/sleep_probe.json` / `stm32_timing.json`. Figures are regenerated by `src/compose_figs.py`, `compose_figs2.py`, `fig_stm32_compare.py`, `fig_sleepwave.py`. SCAPS layer parameters are Table I (SI, as in `perovskite-mapi.def`); spectrum and illumination are Sec. II-D–II-E.

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